FORMAL COMPLETION ALONG A CLOSED SUBSET PRESERVES THE ADIC* AND NOETHERIAN TYPES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one identification of a base change of the completion; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- An algebraic space admits a surjective etale morphism from a disjoint union of affine schemes. Properties of Spaces, Lemma spaces-properties-lemma-cover-by-union-affines, invoked at proof lines 4765-4768. Supplies the covering along which the statement is checked.
- The base change of a completion is the completion along the inverse image. Same chapter, Lemma lemma-map-completions-representable, invoked at proof lines 4770-4772. Identifies the base changes of the completion as completions of the members.
- The affine case: completion along a closed subset of an affine scheme and its types. Same chapter, Lemma lemma-affine-formal-completion-types, invoked at proof lines 4776-4777. Gives both assertions once the hypotheses are transferred.

Source reference. Original lemma 0GBA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3276–3276. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0APV: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3231–3231. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03FX: sources/stacks-chow/tags.txt, source lines 4508–4508. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $T \subset |X|$ be a closed subset. Let $X_{/T}$ be the formal completion of X along T.*

(1) *If $X \setminus T \to X$ is quasi-compact, then $X_{/T}$ is locally adic*.*
(2) *If X is locally Noetherian, then $X_{/T}$ is locally Noetherian.*

Proof. Choose a surjective étale morphism $U \to X$ with $U = \coprod U_i$ a disjoint union of affine schemes, see Properties of Spaces, Lemma 03FX. Let $T_i \subset U_i$ be the inverse image of T. We have $X_{/T} \times_X U_i = (U_i)_{/T_i}$ (Lemma 0APV). Hence $\{(U_i)_{/T_i} \to X_{/T}\}$ is a covering as in Definition 0.1. Moreover, if $X \setminus T \to X$ is quasi-compact, so is $U_i \setminus T_i \to U_i$ and if X is locally Noetherian, so is U_i. Thus the lemma follows from the affine case which is Lemma 0GBA. □